MTech Major Project Selection: Predictive Analytics for Smart City Infrastructure

Direct Recommendation

The strongest subtopic is **urban traffic congestion**, developed not as a simple traffic-prediction application but as a complete **forecast–explain–advise–simulate decision-support system.**

Final project title:

> **Predictive Analytics for Smart City Infrastructure – Multi-Horizon Traffic Congestion Forecasting and Explainable Agentic Signal Advisory Using Spatiotemporal Transformers and RAG**

This keeps the required parent title unchanged, addresses a current operational problem, supports real and recent datasets, and gives meaningful roles to Transformers, explainable AI, retrieval-augmented generation (RAG), an agentic workflow, a large language model (LLM), and a traffic digital twin. The NYC DOT traffic-speed source alone has approximately 112 million records, 13 fields, link-level speed/travel-time observations, and a data date of September 20, 2026, making it suitable for a substantial empirical project rather than a toy demonstration.[^1]

The project should be presented as a **six-month-scale architecture executed through a compressed nine-day engineering sprint.** Its scope remains broad: data engineering, spatiotemporal forecasting, uncertainty estimation, explainability, abnormal-event analysis, RAG-grounded recommendations, multi-agent orchestration, counterfactual simulation, dashboard development, experimentation, and paper preparation. What must not be claimed is physical deployment on a city signal controller; the defensible result is a deployment-ready decision-support prototype validated on real data and in a microscopic traffic simulator.

File Analysis

The supplied project note establishes several non-negotiable requirements:

- Preserve "**Predictive Analytics for Smart City Infrastructure**" as the main title.
- Select one focused infrastructure domain and formulate a scenario-based real-world problem.
- Use current, accessible, real datasets, preferably containing 2024–2026 observations.
- Incorporate Transformers, LLMs, generative AI, RAG, and agentic AI meaningfully.
- Produce implemented results, a research paper, plagiarism-check submission, and a final-review presentation.
- Complete implementation and documentation by October 19, 2026, for the October 21 review.

- Maintain MTech-level technical and research depth despite the compressed execution period.

These conditions eliminate topics whose primary evidence depends on unavailable local sensors, private utility telemetry, long-duration field trials, or hardware installation. They favor a domain with high-frequency public data, measurable targets, well-defined baselines, an accepted simulation environment, and operational documents suitable for RAG. Traffic congestion satisfies all five conditions.

Candidate Subtopics

The original note already identifies traffic, energy, transportation, water, public safety, environment, sustainability, resilience, and system optimization. Additional modern subtopics under the same parent title include:

- Urban flood and storm-water overflow forecasting.
- Water leakage, pipe-burst, and pressure-anomaly prediction.
- Predictive maintenance of roads, bridges, streetlights, and public assets.
- Public-transit delay, bunching, crowding, and demand forecasting.
- Parking occupancy and curb-space demand prediction.
- EV-charging demand and grid-load coordination.
- Building-level energy demand, peak-load, and anomaly forecasting.
- Waste-generation prediction and dynamic collection routing.
- Air-quality hotspot and health-risk forecasting.
- Urban heat-island and heat-stress prediction.
- Emergency-call demand and response-resource positioning.
- Crash-risk and vulnerable-road-user hotspot prediction.
- Citizen-service demand forecasting from 311 complaints.
- Infrastructure cyber-physical anomaly detection.
- Disaster resilience and recovery-time prediction.
- Multimodal mobility optimization involving cars, buses, cycles, and pedestrians.

Ranked Shortlist

Rank	Scenario-based subtopic	Modern problem solved	Data readiness	Research depth	Demonstrability by deadline	Decision
1	Multi-horizon congestion forecasting and signal advisory	Anticipates recurring and event-driven congestion, then tests interventions	Excellent: high-frequency speed, volume, event, weather, and crash sources	Excellent: spatiotemporal learning, XAI, RAG, agents, simulation	Excellent	**Selected**
2	Urban flood-risk prediction and response allocation	Predicts road/waterlogging risk during extreme rainfall	Moderate; labels and drainage telemetry are often fragmented	Excellent	Moderate; event labels and hydraulic validation are difficult	Reserve option
3	Energy peak-load and anomaly management	Forecasts overload and recommends demand response	Moderate; recent aggregate data exist, but city/building telemetry can be restricted	Excellent	Moderate	Strong alternative
4	Air-quality hotspot forecasting and intervention advice	Forecasts PM2.5/NO2 exposure and recommends mitigations	Good via OpenAQ and weather APIs	High	Good, but intervention validation is weaker	Strong alternative
5	Predictive maintenance from 311 and asset records	Prioritizes road/streetlight/tree defects before failure	Good for complaint demand, weaker for verified failure labels	High	Good	Alternative
6	Crash-risk hotspot prediction	Predicts high-risk place/time cells and suggests countermeasures	Good, although the principal NYC crash feed reported an update interruption in 2026	High	Good	Supporting module, not primary topic

Rank	Scenario-based subtopic	Modern problem solved	Data readiness	Research depth	Demonstrability by deadline	Decision
7	Waste-demand forecasting and route optimization	Predicts collection load and optimizes routes	Variable by city	High	Moderate	Alternative
8	Water leak and pipe-burst prediction	Prevents loss and service disruption	Often poor because pressure and failure telemetry are not fully open	Excellent	Low	Not recommended for the deadline

The energy option has genuine recent data: a 2026 repository contains 2024–2025 Linky readings for roughly 100 French households at sub-minute resolution. However, traffic is stronger because it combines a much larger public operational feed with spatial networks, controllable interventions, and visual simulation. Air quality also has a global API covering pollutants such as PM2.5, PM10, SO2, NO2, CO, and O3, but it offers a less direct way to demonstrate that the generated recommendation caused an operational improvement.[2][3]

Selected Scenario

Scenario Definition

A major permitted event begins near a busy urban corridor while rain reduces road capacity and an incident disrupts one connected link. Fixed-time traffic control reacts only after queues have formed. City operators need to know:

1. Which links will become congested 15, 30, and 60 minutes ahead?
2. How severe and uncertain will the congestion be?
3. What traffic, temporal, weather, event, and neighboring-link factors caused the forecast?
4. Which bounded signal-plan adjustment or route advisory should an operator consider?
5. Would that recommendation improve delay, queue length, throughput, stops, and emissions in simulation without causing unsafe or unfair side effects?

The NYC permitted-events dataset contains the event interval, borough, location, event type, and street-closure type for approved events occurring within the next month; it held about 30,700 rows and was updated September 18, 2026. Open-Meteo provides hourly historical weather, with ERA5/ERA5-Land coverage extending to the present and an archive endpoint that accepts coordinates, dates, and selected variables. These sources allow the forecasting model to distinguish routine rush-hour congestion from rain-, event-, and incident-associated disruption.[4][5]

Problem Statement

Current traffic systems commonly detect congestion after it occurs, while many research prototypes stop after predicting speed or assigning a congestion label. The proposed system will ingest current and historical link-level traffic data, create a dynamic road graph, forecast speed and congestion at multiple future horizons, quantify uncertainty, explain each alert, retrieve relevant evidence and operating constraints, generate safe candidate interventions through a bounded multi-agent workflow, and validate those interventions in SUMO before presenting them to a human operator.

Research Questions

- **RQ1:** Does a spatiotemporal Transformer improve 15-, 30-, and 60-minute link-speed forecasts over historical-average, persistence, XGBoost, LSTM/GRU, and non-graph Transformer baselines?
- **RQ2:** Do weather, permitted events, incidents, and neighboring-link states improve performance during non-recurring congestion?
- **RQ3:** Can calibrated uncertainty identify forecasts that should be deferred to an operator rather than acted upon automatically?
- **RQ4:** Can SHAP and attention/neighbor attribution produce stable, operationally meaningful explanations?
- **RQ5:** Does a RAG-grounded agent produce more supported, constraint-compliant recommendations than an ungrounded LLM?
- **RQ6:** Do simulation-approved recommendations reduce delay and queue length relative to fixed-time and actuated baselines without unacceptable effects on side roads, pedestrians, or emergency priority?

Research Hypotheses

- **H1:** Spatial context plus temporal attention will reduce forecasting error, especially at longer horizons.
- **H2:** Event and weather features will give their largest benefit on anomalous days rather than routine periods.
- **H3:** A retrieve–generate–verify agentic pipeline will have fewer unsupported recommendations and constraint violations than direct LLM prompting.
- **H4:** Counterfactual simulation will reject some plausible-sounding LLM recommendations, demonstrating why an LLM must not directly control signals.

Research Gap

Recent traffic-forecasting research confirms that accurate models must capture both temporal dependence and spatial interaction. TEA-GCN addresses the limitation of fixed road graphs by learning changing spatial dependencies and combining local/global temporal attention. A separate 2024 model uses a temporal Transformer and adaptive graph structures to address long-term patterns and heterogeneous dynamic spatial relationships.[6][7]

The opportunity is not merely to introduce another predictor. A 2026 systematic review found that traffic-AI studies remain difficult to compare, with weak external validation, limited transferability, and inadequate reproducibility; explainability appeared in only 23 of 129 model rows in its structured evidence, or 17.8%. The proposed contribution therefore combines reproducible temporal validation, event-specific testing, uncertainty, explanation, grounded operational advice, and simulation-based safety checking.[^8]

LLMs can support transportation planning but should not be treated as numerical forecasters or unconstrained controllers. Recent evaluation found that performance varies with task complexity and recommended RAG, domain adaptation, hybrid systems, and human oversight. A traffic-incident decision-support study has already shown the value of retrieving historical reports and operating manuals before generating decisions. The defensible novelty is therefore the **closed research loop**—predict, explain, retrieve, propose, verify, and defer—not a claim that LLMs alone optimize traffic.[9][10][^11]

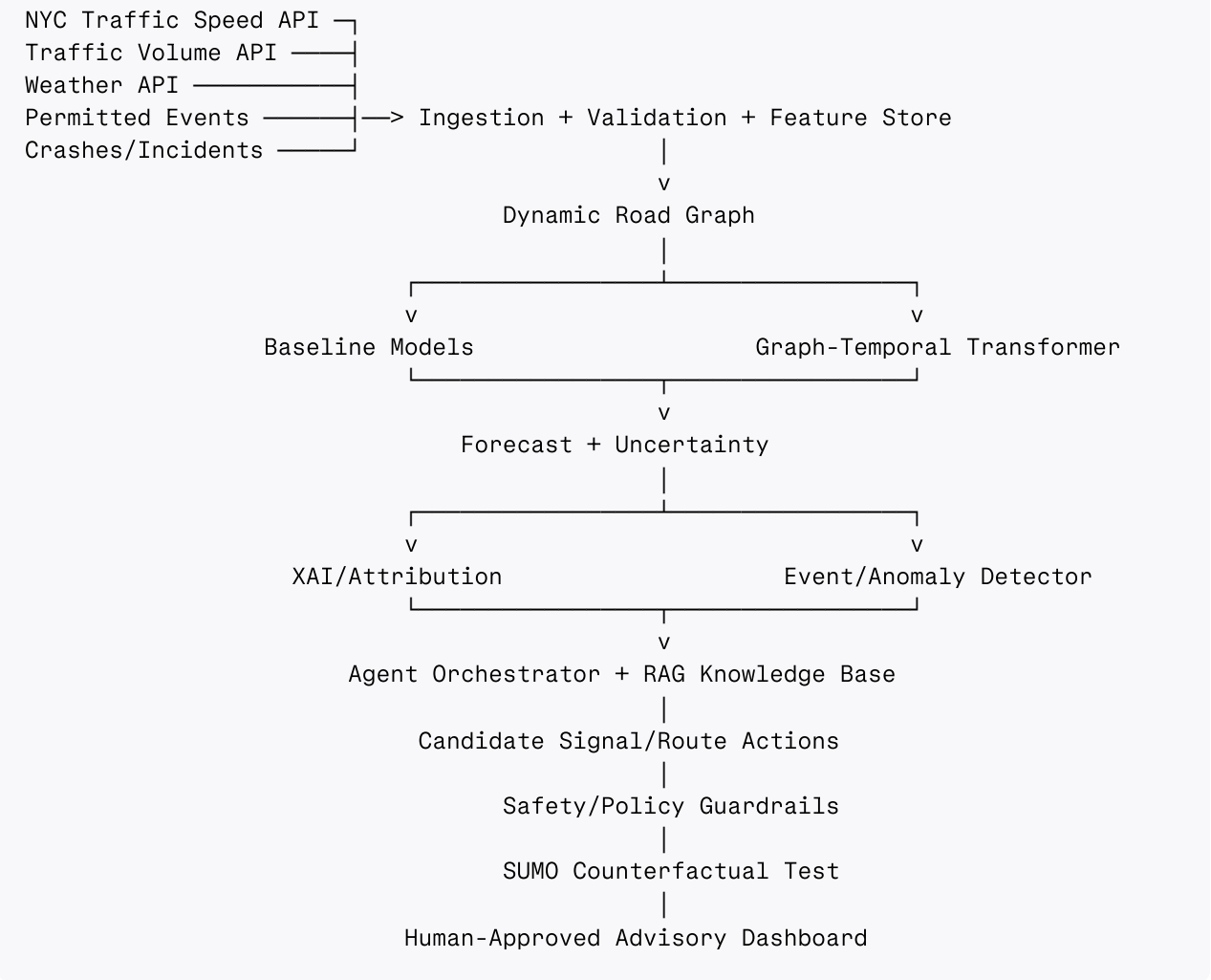
Technical Contribution

The major-project contribution consists of nine integrated work packages:

1. **Automated data lake:** API ingestion, schema validation, data-version manifest, caching, missing-value audit, and reproducible feature generation.
2. **Road-graph construction:** Nodes represent observed road links or sensors; weighted edges encode physical adjacency, distance, direction, or data-driven correlation.
3. **Multi-horizon forecasting:** A graph-aware Transformer predicts link speed and congestion for 15, 30, and 60 minutes ahead.
4. **Baselines and ablations:** Historical average, persistence, XGBoost, LSTM/GRU, and temporal Transformer are evaluated against the proposed model; feature families and graph components are removed systematically.
5. **Uncertainty and anomaly layer:** Conformal intervals or quantile loss estimate risk; an event detector marks non-recurring patterns and out-of-distribution inputs.
6. **Explainability layer:** SHAP explains tabular and tree baselines, while temporal attention and neighboring-link attribution explain the deep model. SHAP is based on game-theoretic Shapley values and supports model-agnostic explanations; TreeExplainer provides a fast exact method for supported tree ensembles.[12][13]
7. **RAG knowledge layer:** A vector index stores signal-control rules, traffic-management procedures, project assumptions, data dictionaries, model cards, and prior scenario reports. FAISS supplies efficient dense-vector similarity search and Python/GPU support. [^14]
8. **Bounded agentic layer:** Specialized agents monitor data, call the forecast model, inspect uncertainty and explanations, retrieve evidence, propose candidate actions, check policy/safety constraints, call simulation, and prepare a cited operator report.
9. **Digital-twin validation and dashboard:** SUMO/TraCI tests candidate interventions; a Streamlit or FastAPI/React dashboard maps predictions, explanations, confidence, retrieved evidence, recommendations, and before/after simulation metrics.

This is a major project because the output is an integrated decision-support platform with a formal experimental program—not a single notebook or one trained model.

System Architecture



The LLM is intentionally downstream of the numerical model. It translates machine outputs and retrieved evidence into an operator-readable advisory; it does not invent forecasts, modify model results, or bypass constraints.

Data Strategy

Core Sources

Dataset	Intended use	Evidence of suitability	Important limitation
NYC DOT Traffic Speeds, ID `i4gi-tjb9`	Primary target: speed/travel time by road link and timestamp	About 112 million rows, 13 columns, average speed between endpoints, data dated September 20, 2026[^1]	Verify sampling continuity and retired links before modeling

Dataset	Intended use	Evidence of suitability	Important limitation
NYC Automated Traffic Volume Counts, ID `7ym2-wayt`	Traffic-demand covariate and historical validation	About 1.88 million rows; 15-minute volume, segment, geometry, street, and direction; updated March 20, 2026[^15]	Intermittent samples; locations and days vary by year
NYC Permitted Event Information, ID `tvpp-9vvx`	Known event and road-closure features	About 30,700 rows; next-month approved events, times, location, type, and street-closure field; updated September 18, 2026[^5]	Not all events affect traffic; geocoding may be required
Open-Meteo Historical/Forecast API	Rain, temperature, wind, visibility-related variables	Hourly global historical data and forecast-compatible APIs; no-key access is available for non-commercial use[4][16]	Reanalysis grid is coarser than a street sensor
NYC Motor Vehicle Collisions	Incident and crash-risk context	Event date/time, location, injuries, fatalities, and contributing information are available[^17]	Main feed reported a temporary update interruption; treat as historical context, not the sole live input
NYC 311 service requests	Road-condition and citizen-observation features	Base data cover 2010–present and are automatically updated daily; several views were current in October 2026[18][19]	Complaints reflect reporting behavior, not direct sensor truth

Data Discipline

The code should pull bounded date ranges through Socrata queries rather than attempting to download all 112 million speed rows. Store raw API responses, a data dictionary, retrieval time, endpoint, query, checksum, license note, and processed-file version. Use 2024–2025 for training/validation where sufficiently complete, hold out the latest available 2026 interval for testing, and never randomly shuffle a time series.

Before freezing the topic, run a two-hour data audit:

- Confirm that 2024, 2025, and 2026 traffic-speed records are actually returned by the API.
- Count unique links and timestamps by month.
- Quantify missingness, duplicate records, zero speeds, impossible values, and sensor outages.
- Select a corridor/subnetwork with consistent coverage; retain citywide ingestion and mapping as the platform layer.
- Join permitted events and weather only after validating timestamps and spatial mapping.
- Record the exact cutoff so later API changes do not invalidate reported results.

Using a corridor for model training is not a reduction to a toy project: the architecture remains city-scale, while a controlled empirical test makes the results reproducible and computationally defensible. The paper should explicitly distinguish **platform scope** (city-scale) from **experimental scope** (selected high-quality network).

Forecasting Method

Target Definition

Predict continuous link speed at horizons of 15, 30, and 60 minutes. Derive a secondary congestion state—free, moderate, heavy, severe—from the ratio of predicted speed to a link-specific free-flow baseline, with thresholds declared before testing. Continuous prediction prevents arbitrary labels from becoming the only evidence.

For link i and horizon h, a normalized congestion index can be defined as:

$$CI_{i,t+h} = 1 - \frac{\hat{v}_{i,t+h}}{v_i^{free}}$$

where $\hat{v}_{i,t+h}$ is forecast speed and v_i^{free} is a robust off-peak speed estimate. Cap or carefully handle anomalous ratios, and document how the free-flow estimate is computed.

Proposed Model

Use a **Graph-Temporal Fusion Transformer** or similarly named architecture with:

- Temporal embeddings for minute, hour, weekday, weekend, and holiday.
- Lagged speed, travel time, volume, rolling mean, rolling variance, and trend.
- Weather, event, closure, incident, and 311-derived contextual features.
- Static link embeddings for borough, direction, length, and road class when available.
- Graph message passing or graph attention for upstream/downstream interactions.
- Temporal self-attention for short- and long-range patterns.
- Multi-task heads for 15-, 30-, and 60-minute speed.
- Quantile heads or conformal calibration for prediction intervals.

Transformer and adaptive-graph models are appropriate because recent work identifies long temporal dependencies and changing spatial relationships as central challenges. Do not claim architectural novelty merely from combining named layers; frame the main novelty as operational integration, robust evaluation, uncertainty, explainability, and verified agentic recommendations.[7][6]

Baselines

- Last observed value (persistence).
- Historical mean by link, weekday, and time slot.
- XGBoost or LightGBM on engineered lags.
- LSTM or GRU.
- Temporal Transformer without graph information.
- Proposed graph-temporal Transformer.

A complex deep model must earn its inclusion. Some research reports that simple linear forecasting can outperform complex GNNs in accuracy and training efficiency on selected benchmarks. Therefore, reporting all baselines—even when the proposed model loses on a horizon—is academically stronger than hiding inconvenient results.[^20]

Agentic RAG Design

Knowledge Base

Index only traceable documents:

- Dataset dictionaries and metadata.
- Traffic-control and incident-response manuals that permit reuse.
- SUMO/TraCI documentation.
- Model card, feature definitions, known limitations, and thresholds.
- Approved intervention templates and safety constraints.
- Generated historical scenario reports with measured simulation outcomes.

Every advisory should show retrieved source chunks, source name, and retrieval score. An unsupported recommendation should fail validation rather than be displayed as established fact.

Agent Roles

Agent	Allowed responsibility	Prohibited behavior
Data Quality Agent	Check freshness, missingness, drift, and sensor anomalies	Impute silently or declare bad data valid
Forecast Agent	Call the trained model and return predictions/intervals	Ask the LLM to calculate the numerical forecast
Explanation Agent	Produce SHAP/attention/neighbor evidence	Treat attention alone as causal proof
Event Agent	Correlate weather, closures, events, and anomalies	Declare causation from correlation
Retrieval Agent	Retrieve operational evidence and constraints	Use uncited general memory when the knowledge base is available
Planner Agent	Generate bounded candidate actions	Directly alter real signals
Safety Agent	Reject actions violating phase, pedestrian, emergency, or confidence rules	Override hard constraints
Simulation Agent	Run SUMO counterfactuals and compare metrics	Replace field validation with unqualified claims
Report Agent	Produce a cited operator advisory	Change forecast/simulation numbers

The orchestration state machine should be explicit: `INGEST -> VALIDATE -> FORECAST -> EXPLAIN -> RETRIEVE -> PROPOSE -> CHECK -> SIMULATE -> APPROVE/REJECT/DEFER -> REPORT`.

This is genuine agentic design because each transition is tool- and condition-driven, not simply a sequence of chat prompts.

Simulation Design

SUMO is a microscopic, space-continuous, time-discrete traffic simulator. TraCI allows a Python controller to read simulation state and change signal behavior online through a client/server interface. SUMO supports fixed, actuated, and externally controlled signals; a common adaptive pattern lets SUMO retain yellow/all-red handling while a controller decides when a green phase ends.[21][22][23][24]

Compared Policies

- Fixed-time control.
- SUMO actuated control.
- Forecast-informed rule-based policy.
- Agent-proposed policy after safety filtering.
- Optional emergency-priority scenario; SUMO includes configurable emergency-vehicle behavior.[^25]

Intervention Space

Keep actions bounded and interpretable:

- Extend or shorten the current green within declared minimum/maximum limits.
- Select among pre-approved phase plans.
- Apply corridor offsets within safe bounds.
- Recommend diversion or operator messaging.
- Defer action when uncertainty, data quality, or simulation outcome is unacceptable.

Simulation Metrics

- Mean and 95th-percentile travel time.
- Mean delay and stopped delay.
- Mean and maximum queue length.
- Throughput and completed trips.
- Number of stops.
- Side-road and pedestrian waiting time.
- Emergency-vehicle travel time in the priority scenario.
- Fuel/emission proxy if SUMO output is configured.
- Safety/constraint violations.

Run multiple random seeds and report mean, dispersion, and paired differences. Avoid presenting a single successful animation as proof.

Evaluation Protocol

Forecast Metrics

- MAE, RMSE, MAPE or sMAPE, and R^2 by horizon.
- Per-link and aggregated metrics.
- Peak versus off-peak performance.
- Routine versus event/anomaly-day performance.
- Prediction-interval coverage and average interval width.
- Inference latency and resource use.

Experimental Validity

Use rolling-origin or chronological train/validation/test splits. Include an event-held-out test, a missing-sensor stress test, and at least one transfer test to unseen links or a different corridor if the data permit. Report ablations for graph edges, weather, events, volume, uncertainty calibration, RAG, safety verification, and simulation verification.

RAG and Agent Metrics

Create a small, manually checked evaluation set of scenario questions and required evidence. Measure:

- Retrieval Recall@k and Mean Reciprocal Rank.
- Citation correctness and evidence coverage.
- Groundedness or unsupported-claim rate.
- Constraint-violation rate.
- Correct tool-selection rate.
- Recommendation acceptance/rejection/defer rate.
- End-to-end latency.
- Difference between direct-LLM, RAG-only, and agentic-RAG outputs.

Acceptance Criteria

The final paper should not promise predetermined accuracy. Instead, define success before training:

- Proposed forecast model beats persistence and historical average at all reported horizons.
- Prediction intervals achieve near-nominal coverage after calibration.

- RAG agent materially reduces unsupported recommendations and policy violations versus direct prompting.
- At least one scenario shows statistically reported simulation improvement over fixed time without worsening a protected metric beyond a stated tolerance.
- Failure cases and rejected recommendations are included.

Nine-Day Execution Plan

The working dates are October 11–19, 2026, followed by review preparation for October 21. Parallel execution with coding agents is useful, but every generated module must pass tests, data assertions, and manual review.

Date	Primary output	Non-negotiable completion gate
Oct 11	Topic freeze, title, scenario, repository, source manifest	Dataset API returns verified 2024–2026 samples; target and corridor are frozen
Oct 12	Ingestion and EDA pipeline	Raw/processed versioning, missingness report, map, and leakage-safe split
Oct 13	Baselines	Persistence, historical average, XGBoost, and LSTM/GRU metrics saved
Oct 14	Proposed Transformer	Multi-horizon training, checkpointing, metrics, learning curves
Oct 15	XAI and uncertainty	SHAP/attribution plots, calibrated intervals, error slices, ablations started
Oct 16	RAG and agents	Indexed knowledge base, orchestration graph, citations, guardrails, agent tests
Oct 17	SUMO digital twin	Fixed, actuated, and proposed policies run over multiple seeds; metrics exported
Oct 18	Dashboard and full experiments	End-to-end demo, maps, comparisons, ablations, scenario replay, failure cases
Oct 19	Paper and reproducibility freeze	Paper, abstract, results tables, README, environment lock, final code/tag complete
Oct 20	Presentation rehearsal	Slides, architecture figure, demo video backup, viva questions, plagiarism revisions
Oct 21	Review	Live demo plus offline fallback, evidence package, paper, and results

Parallel Workstreams

- **Agent A:** Data connectors, schema tests, feature store.
- **Agent B:** Baselines, Transformer, uncertainty.
- **Agent C:** Explainability, evaluation, plots.
- **Agent D:** RAG corpus, retrieval tests, orchestration.
- **Agent E:** SUMO network, policies, scenario runner.

- **Agent F:** Dashboard, documentation, paper scaffolding.

Parallelism must not create six incompatible prototypes. Freeze shared schemas, configuration files, random seeds, metric names, and artifact directories on the first day.

Repository Structure

```
smart-city-traffic-project/
├── configs/
├── data/
│   ├── raw_manifest/
│   ├── interim/
│   └── processed/
├── docs/knowledge_base/
├── src/
│   ├── ingestion/
│   ├── validation/
│   ├── features/
│   ├── graph/
│   ├── models/
│   ├── uncertainty/
│   ├── explainability/
│   ├── rag/
│   ├── agents/
│   ├── simulation/
│   └── dashboard/
├── tests/
├── experiments/
├── artifacts/
│   ├── models/
│   ├── metrics/
│   ├── figures/
│   └── scenario_reports/
├── paper/
├── presentation/
├── Dockerfile
├── requirements.lock
└── README.md
```

Deliverables

The completed submission should include:

- Reproducible source code and environment specification.
- API-driven data ingestion with provenance manifest.
- Data-quality report and exploratory analysis.
- Trained baseline and proposed models.
- Saved model checkpoints and configuration.
- Metrics by horizon, scenario, link, and model.
- Ablation and stress-test tables.

- Explainability and uncertainty evidence.
- RAG corpus and retrieval benchmark.
- Agent traces showing approvals, rejections, and deferrals.
- SUMO scenario files and multi-seed outputs.
- Interactive operator dashboard.
- Research paper and plagiarism-check-ready manuscript.
- Final presentation, architecture diagram, and offline demo recording.
- Ethical, limitation, and reproducibility statements.

Paper Blueprint

Proposed Abstract

Urban traffic congestion is a dynamic smart-city infrastructure problem affected by temporal demand, neighboring road conditions, weather, planned events, and incidents. This project proposes an end-to-end decision-support framework that forecasts link-level traffic speed and congestion at 15-, 30-, and 60-minute horizons using a graph-aware spatiotemporal Transformer. The framework integrates calibrated uncertainty and explainable AI to identify the temporal, contextual, and spatial factors behind each forecast. A retrieval-augmented multi-agent layer uses traceable operating knowledge to generate bounded signal and route advisories, while safety rules and SUMO counterfactual simulation verify candidate actions before human review. The system is evaluated using recent public traffic, volume, weather, event, and incident data with chronological validation, baseline comparisons, ablation studies, anomalous-event tests, retrieval metrics, and multi-seed simulation experiments. The intended contribution is a reproducible forecast–explain–advise–simulate loop that connects predictive accuracy with transparent and operationally constrained smart-city decision support.

The abstract must be revised after experiments to insert only measured results. Do not invent accuracy, percentage improvement, latency, or emissions reduction.

Chapter Structure

1. Introduction and motivation.
2. Problem definition, scenario, objectives, and contributions.
3. Literature review: statistical, ML, deep temporal, graph, Transformer, XAI, RAG/LLM, agentic, and simulation methods.
4. Research gaps and hypotheses.
5. Data sources, governance, cleaning, and exploratory analysis.
6. Proposed architecture and mathematical formulation.
7. Implementation: prediction, uncertainty, XAI, RAG, agents, simulation, dashboard.
8. Experimental design and baselines.
9. Results: forecasting, ablations, uncertainty, RAG, agents, and simulation.

10. Discussion, validity threats, ethical issues, and limitations.
11. Conclusion and future work.

Viva Positioning

Use the following concise defense:

> "The project is not an LLM traffic chatbot. Its core is a reproducible spatiotemporal forecasting and decision-validation pipeline. The Transformer forecasts congestion; uncertainty identifies when not to trust it; explainability shows the evidence; RAG grounds operational recommendations; specialized agents enforce a verifiable workflow; and SUMO evaluates each intervention before a human sees it."

Likely panel questions should be answered as follows:

- **Why traffic?** It offers recent high-frequency operational data, a graph-structured physical system, measurable prediction targets, and controllable simulated interventions.
- **Why Transformer?** It models longer temporal interactions; the graph component adds neighboring-link dependence. Both must be justified empirically against simpler baselines.
- **Why not let an LLM predict speed?** LLMs are used for evidence-grounded explanation and planning, not numerical time-series forecasting.
- **What is novel?** The closed loop joining multi-horizon prediction, calibrated uncertainty, explanation, RAG-grounded agents, hard guardrails, and counterfactual simulation under reproducible evaluation.
- **Is this deployed in NYC?** No. It is a research decision-support prototype using NYC open data and a SUMO digital twin; no real controller is modified.
- **How is hallucination controlled?** Retrieved evidence, tool-only numeric outputs, structured schemas, hard constraints, citation checks, simulation verification, and human approval.
- **Why not a larger geographic test?** The platform is designed for city-scale ingestion; experiments use a high-quality subnetwork to ensure temporal completeness, reproducibility, and defensible computation.

Risks and Mitigations

Risk	Consequence	Mitigation
Latest speed API records are sparse or inconsistent	Weak 2026 test set	Audit immediately; freeze the latest complete interval and disclose cutoff
Volume data are intermittent	Poor direct alignment	Use as contextual feature where matched, not mandatory target; report missingness
Event locations do not geocode cleanly	Noisy context feature	Use borough/corridor matching plus a manually validated event subset
Deep model underperforms XGBoost	Weak novelty claim	Report honestly; emphasize closed-loop system and analyze where complexity helps

Risk	Consequence	Mitigation
GPU/time constraints	Incomplete tuning	Train on a consistent corridor, use mixed precision, early stopping, small search, fixed seeds
LLM gives unsafe advice	Invalid demonstration	Bounded actions, schemas, hard rules, simulation gate, and human approval
SUMO network calibration is weak	Overstated intervention benefit	Calibrate demand from observed volumes; present simulation as counterfactual evidence, not field proof
API changes after submission	Non-reproducible results	Cache immutable extracts and store queries, checksums, and timestamps
Paper-writing time collapses	Incomplete academic output	Write methods and literature sections from day one; generate tables/figures automatically

Future Scope

- Transfer learning across cities and unseen road networks.
- Federated or privacy-preserving learning for sensitive mobility streams.
- Continual learning with formal drift detection.
- Causal evaluation of interventions rather than correlational attribution.
- Multi-objective optimization across vehicles, buses, pedestrians, cyclists, emissions, and equity.
- Vehicle-to-everything and connected-intersection inputs.
- Real-time digital twins with live calibration.
- Human factors study of operator trust and explanation usefulness.
- Formal verification of agent policies and traffic-signal constraints.
- Field pilot with an authorized traffic-management agency.

Final Decision

Select **traffic congestion forecasting with explainable, agentic signal advisory and digital-twin validation.** It is the best match for the preserved parent title, the strongest combination of current public data and measurable impact, and the most convincing way to integrate Transformers, XAI, RAG, LLMs, generative AI, agents, and simulation without using those technologies as decorative labels.

The final title should remain:

> **Predictive Analytics for Smart City Infrastructure – Multi-Horizon Traffic Congestion Forecasting and Explainable Agentic Signal Advisory Using Spatiotemporal Transformers and RAG**

The first implementation decision is not model selection; it is the October 11 data audit. If the API audit verifies adequate 2024–2026 coverage on a consistent road subnetwork, lock this title and begin the architecture above. If it fails, retain the exact title and method but use the latest

complete traffic period for training and designate 2026 weather/event data for live scenario inference, documenting the temporal mismatch rather than fabricating “current” records.

References

1. DOT Traffic Speeds | NYC Open Data
2. About the API - OpenAQ Docs - Discover the OpenAQ API, which offers open access to global air quality data based on REST principle...
3. Recherche Data Gouv - In the field of modeling electricity consumption behavior among French households, access to observa...
4. Historical Weather API - Historical weather data from 1940 onwards with weather records dating back to 1940 and hourly res...
5. NYC Permitted Event Information | NYC Open Data
6. TEA-GCN: Transformer-Enhanced Adaptive Graph Convolutional Network for Traffic Flow Forecasting - Traffic flow forecasting is crucial for improving urban traffic management and reducing resource con...
7. Deep transformer-based heterogeneous spatiotemporal ... - Accurate geographical traffic forecasting plays a critical role in urban transportation planning, tr...
8. Artificial Intelligence and Machine Learning for Road Traffic Congestion Prediction and Forecasting: A Systematic Review of Methods, Validation, Explainability, and Reproducibility - Artificial intelligence (AI) and machine learning (ML) are increasingly applied to road traffic cong...
9. arXiv:2503.12085v1 [cs.AI] 15 Mar 2025
10. Beyond words: evaluating large language models in transportation planning - The rapid advancement of Generative Artificial Intelligence (GenAI) in 2023 has catalyzed transforma...
11. Beyond words: evaluating large language models in transportation planning - The rapid advancement of Generative Artificial Intelligence (GenAI) in 2023 has catalyzed transforma...
12. shap.readthedocs.io › en › latestWelcome to the SHAP documentation
13. shap.readthedocs.io › shapshap.TreeExplainer — SHAP latest documentation
14. Welcome to Faiss Documentation — Faiss documentation - Faiss is a library for efficient similarity search and clustering of dense vectors. It contains algo...
15. Automated Traffic Volume Counts - NYC Open Data - New York City Department of Transportation (NYC DOT) uses Automated Traffic Recorders (ATR) to colle...
16. Open-Meteo Weather API - GitHub - Free Weather Forecast API for non-commercial use. Contribute to open-meteo/open-meteo development by...
17. Motor Vehicle Collisions - Crashes - NYC Open Data
18. 311 DSNY | NYC Open Data
19. cb 11 311 | NYC Open Data

20. Under review as a conference paper at ICLR 2024
21. TraCI4Traffic Lights - SUMO Documentation
22. TraCI - SUMO Documentation
23. Options
24. Traffic Lights - SUMO Documentation - TraCI provides various functions for controlling traffic lights. And the basic approaches are outlin...
25. Emergency - SUMO Documentation